NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES (FAST-NUCES)

Department of Software Engineering — Semester Project

Cast-Ware: AI-Powered Phishing URL & Email Detector

Project Category: R&D (AI / Cyber Security / Web Tool) Proposed Supervisor: Dr./Mr./Ms. Sarah Abid
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1. Abstract / Executive Summary

Phishing remains the primary initial access vector in modern cyberattacks, exploiting human trust via deceptive Uniform Resource Locators (URLs) and socially engineered emails. Traditional signature and blacklist mechanisms (e.g., Google Safe Browsing) suffer from severe latency against zero-day phishing sites and ephemeral domains. This project proposes Cast-Ware, an end-to-end, Explainable AI (XAI) web platform where users paste a suspicious URL or email body and immediately receive a risk verdict (Phishing vs. Safe), a calibrated confidence score, key threat indicators, and a human-readable explanation. By combining engineered lexical/structural URL features using gradient-boosted decision trees (Random Forest, XGBoost, LightGBM) with a fine-tuned transformer model (DistilBERT) for semantic text analysis, Cast-Ware ensures high resilience against obfuscated attacks while educating users via SHAP-driven feature attribution.

2. Problem Statement & Motivation

Modern threat actors deploy ephemeral phishing kits, subdomain chaining, homograph substitutions, and LLM-generated persuasive emails that bypass static rule filters before blacklists can index them. Furthermore, existing consumer and analyst tools operate as "black boxes"—returning a binary label without explaining why an input is dangerous. Non-technical users often ignore warnings they do not understand, while junior SOC analysts waste valuable triage time manually inspecting URL tokens and email headers. There is a pressing need for a fast, dataset-driven machine learning system capable of evaluating unindexed URLs and email text in real time while transparently surfacing the underlying indicators of compromise (IoCs).

3. Project Objectives

1. Lexical & Structural URL Feature Extractor: Engineer an automated pipeline extracting 30+ intrinsic URL attributes (length ratios, special characters, subdomain count, Shannon entropy, HTTPS misuse, IP presence, and suspicious keywords).
2. High-Accuracy Ensemble & Deep Learning Models: Train, tune, and benchmark strong tabular baselines (Random Forest, XGBoost, LightGBM) alongside a fine-tuned DistilBERT model for semantic email and sequence classification.
3. Explainable AI (XAI) Integration: Utilize SHAP (SHapley Additive exPlanations) and attention attribution to translate complex model weights into clear, user-friendly risk explanations and visual feature contribution charts.
4. Interactive Web Application: Build a responsive web interface (React + FastAPI) allowing seamless submission of URLs or email text with sub-second inference latency.

4. Literature Review & Existing Systems

- Blacklist Services (PhishTank, OpenPhish, Google Safe Browsing): Provide near-zero false positives on known threats but fail entirely against newly registered zero-hour domains until manual verification occurs.
- VirusTotal: Aggregates 70+ third-party domain scanners but relies primarily on historical domain reputation rather than deep semantic evaluation of raw email text or novel lexical URL structures.
- Academic Benchmarks (PhiUSIIL, 2024): Recent studies confirm that tree-based ensembles trained on lexical URL properties achieve $> 98\%$ accuracy on large corpora (100k–700k samples). Cast-Ware bridges the gap between academic models and practical usability by coupling dual URL/Email modalities with real-time SHAP explainability.

5. Proposed Methodology & Architecture

5.1 5.1 Datasets

The project leverages clean, high-volume public corpora ideal for robust training within an academic timeline:

- URL Corpora: PhiUSIIL Phishing URL Dataset (235k samples), live verified PhishTank feeds, and Kaggle Malicious URLs Dataset (651k samples spanning benign, phishing, defacement, and malware URLs).
- Email Corpora: Enron Email Dataset paired with the Nazario Phishing Corpus and Kaggle Phishing Email collections (18k+ labeled emails).

5.2 5.2 Feature Engineering (URL Pipeline)

Given a raw URL string u, our feature extraction module maps $u \mapsto \mathbf{x} \in \mathbb{R}^d$ without requiring external third-party WHOIS lookups (ensuring zero network latency and privacy). Extracted features include:

- Lexical Counts & Ratios: Total URL length L_{url}, hostname length L_{host}, path length, count of special symbols (`@`, `-`, `_`, `?`, `=`, `%`), digit-to-letter ratio, and subdomain depth (n_{sub}).
- Shannon Entropy (H): Quantifies character randomness to detect algorithmically generated domains (DGAs) and obfuscated query strings:

$$H(X) = -\sum_{i=1}^{k} P(x_i) \log_2 P(x_i)$$

- Binary Heuristic Flags: Raw IPv4/IPv6 address in hostname, deceptive `https` token embedded inside the domain/path, URL shortening service detection, and presence of high-risk social engineering tokens (`login`, `verify`, `account`, `update`, `secure`, `bank`).

5.3 5.3 Classification & Explainability Pipeline

1. Tabular URL Classifier: Feature vector $\mathbf{x}$ is evaluated by **Random Forest**, **XGBoost**, and **LightGBM** models. Hyperparameters are optimized via Optuna using Stratified 5-Fold Cross-Validation.
2. Semantic Email & Sequence Classifier: Email bodies (and raw URL text) are tokenized via WordPiece and passed into a fine-tuned **DistilBERT** architecture to capture contextual urgency, financial coercion, and spoofing patterns.
3. XAI Risk Synthesizer: For URLs, TreeSHAP computes exact Shapley values ϕ_i for each feature, ranking the top k contributors pushing the base probability toward phishing. For emails, token-level attributions highlight suspicious phrases. A rule-guided template engine converts these top weights into natural language explanations.

6. Project Scope & Deliverables

- In Scope: Lexical/structural URL feature extraction, ML ensemble training (RF/XGBoost/LightGBM), DistilBERT fine-tuning for email text, SHAP-based explanation generation, FastAPI backend, and React web dashboard.
- Out of Scope: Dynamic client-side webpage rendering (headless browser sandboxing) and enterprise mail-server (SMTP) gateway plugins, keeping the project computationally lightweight and achievable within two semesters.

7. Tools, Libraries & Technologies

Programming Languages:	Python 3.11 (ML/Backend), TypeScript / JavaScript (Frontend)
Machine Learning & NLP:	Scikit-Learn, XGBoost, LightGBM, PyTorch, HuggingFace Transformers (DistilBERT)
Explainable AI & Data:	SHAP (TreeSHAP / PartitionSHAP), LIME, Pandas, NumPy, BeautifulSoup4, tldextract
Web Stack & DevOps:	FastAPI, Uvicorn, React.js, Tailwind CSS, Docker, Git/GitHub

8. Work Breakdown Structure & Tentative Timeline

Phase	Key Tasks & Milestones	Timeline	Lead
Phase 1	Literature review, dataset aggregation (PhiUSIIL, PhishTank, Enron), data cleaning	Sep – Oct 2026	All Members
Phase 2	URL lexical feature engineering pipeline (30+ features) & exploratory data analysis	Nov 2026	Member 1, 2
Phase 3	Training & benchmarking Random Forest, XGBoost, LightGBM (FYP-I Defense)	Dec 2026	Member 1, 3
Phase 4	Fine-tuning DistilBERT on phishing email corpora & sequence evaluation	Jan – Feb 2027	Member 2, 3
Phase 5	SHAP explainability integration, natural language rule synthesizer, FastAPI backend	Mar 2027	Member 1, 2
Phase 6	React frontend dashboard, end-to-end latency testing, ROC-AUC benchmarks, Thesis	Apr – May 2027	All Members

9. References

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